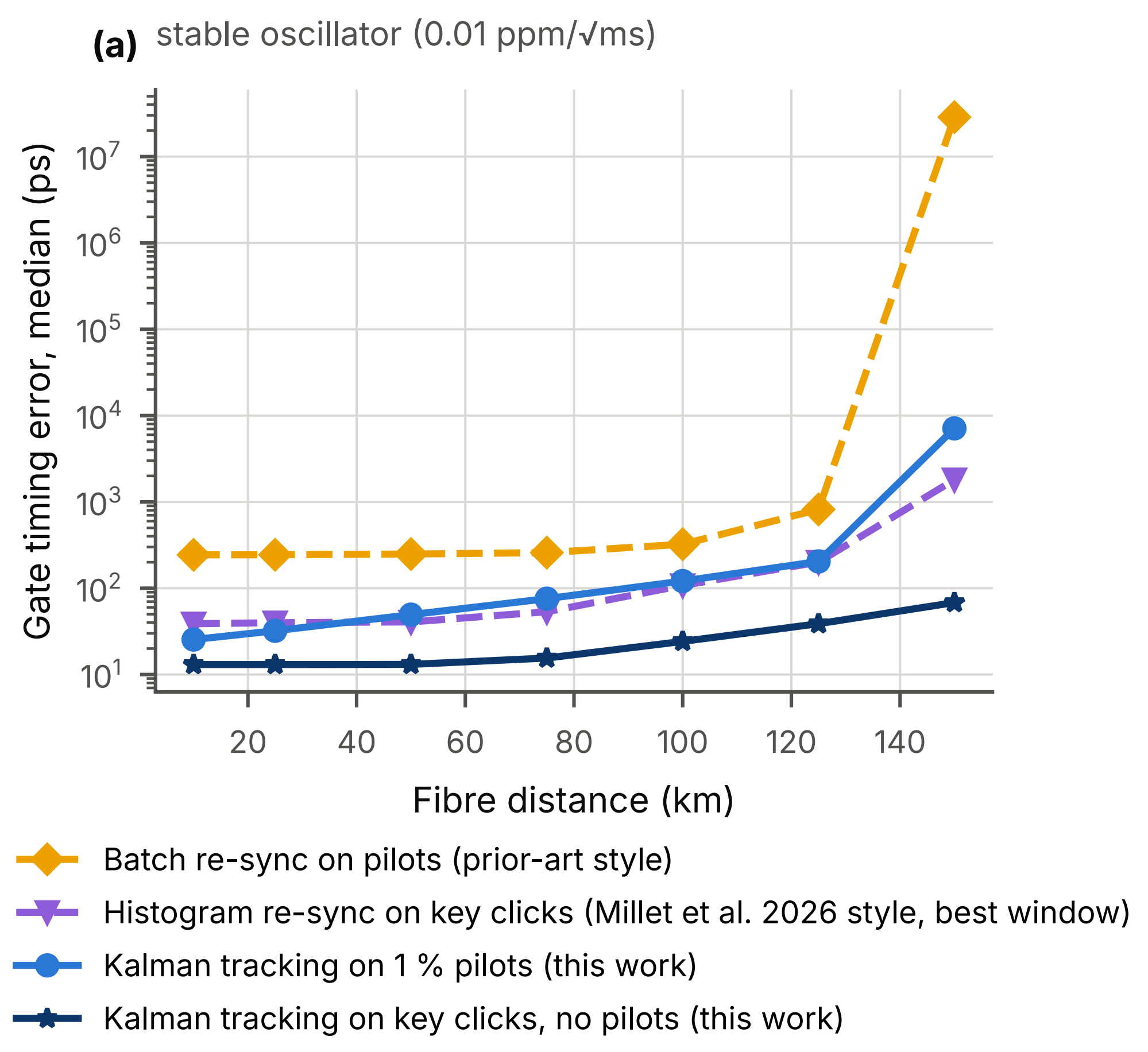
(a) stable oscillator (0.01 ppm/√ms)
Gate timing error, median (ps)
10^7
10^6
10^5
10^4
10^3
10^2
10^1
20
40
60
80
100
120
140
Fibre distance (km)
Batch re-sync on pilots (prior-art style)
Histogram re-sync on key clicks (Millet et al. 2026 style, best window)
Kalman tracking on 1 % pilots (this work)
Kalman tracking on key clicks, no pilots (this work)